



# Capstone DVA Group – 18





# Team Members

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# Problem Statement

**Title:** Operational Flight Delay Analysis

**Content:**

- Why flight delays matter (cost, congestion, passenger impact)
- Objective:

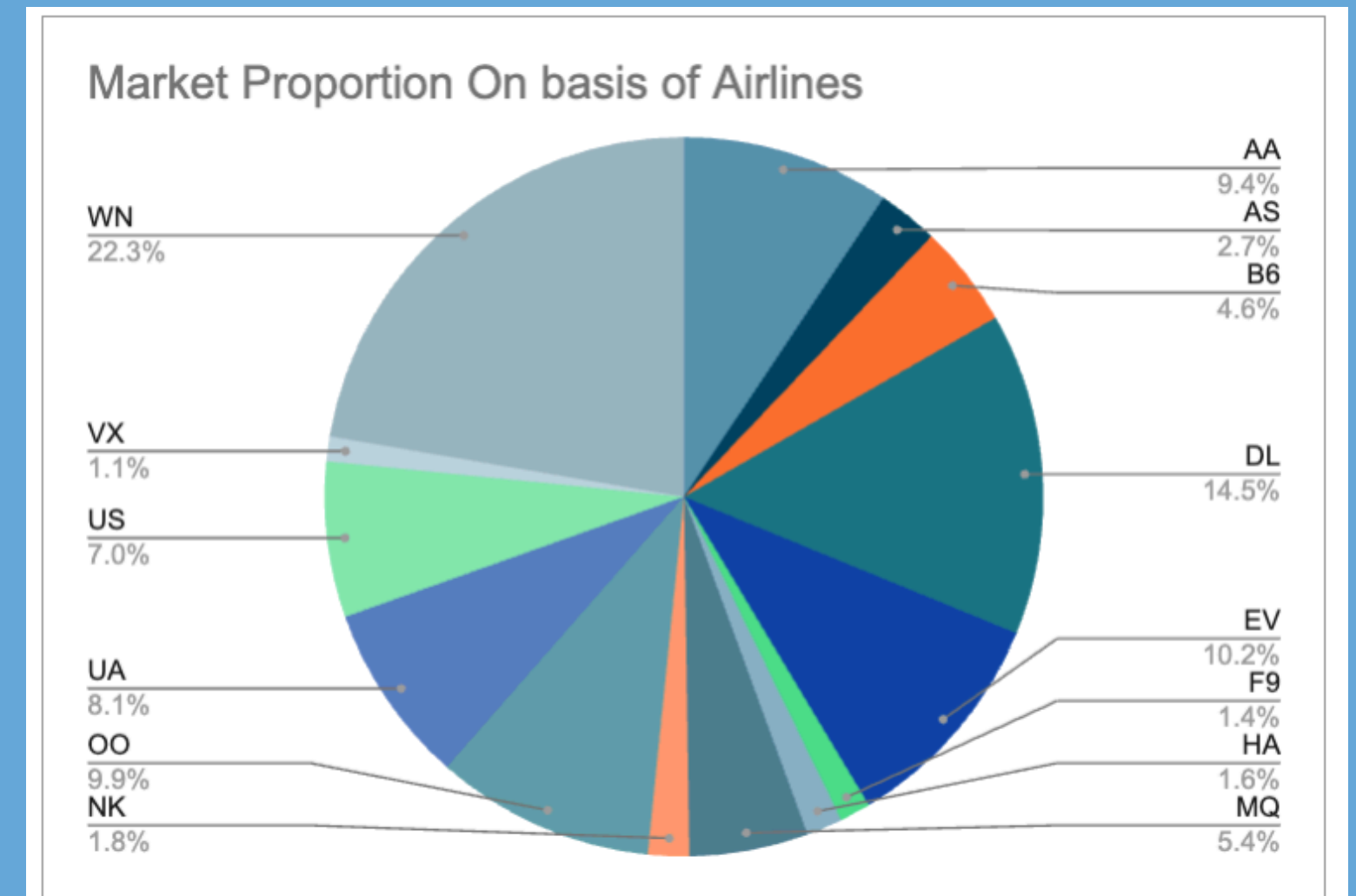
Identify structural delay patterns across airlines, weekdays, and airports to enable targeted operational improvement.

# Data Cleaning Framework

- Duplicate removal
- Tail number validation
- Departure time cleaning
- Taxi-out validation
- Elapsed time validation
- Cancelled & diverted filtered

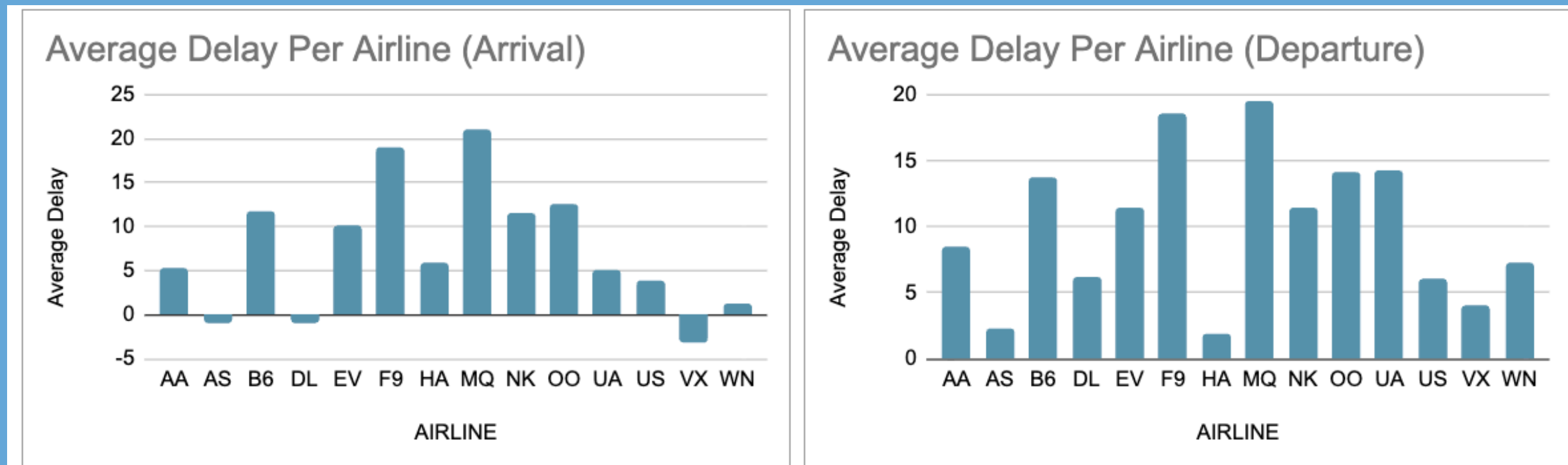
# KPI Framework

- Average Departure Delay
- Average Arrival Delay
- Delay Rate (%)
- Major Delay Rate
- Total Flights per Airline
- Per Day Average Delay



# Airline Performance Findings

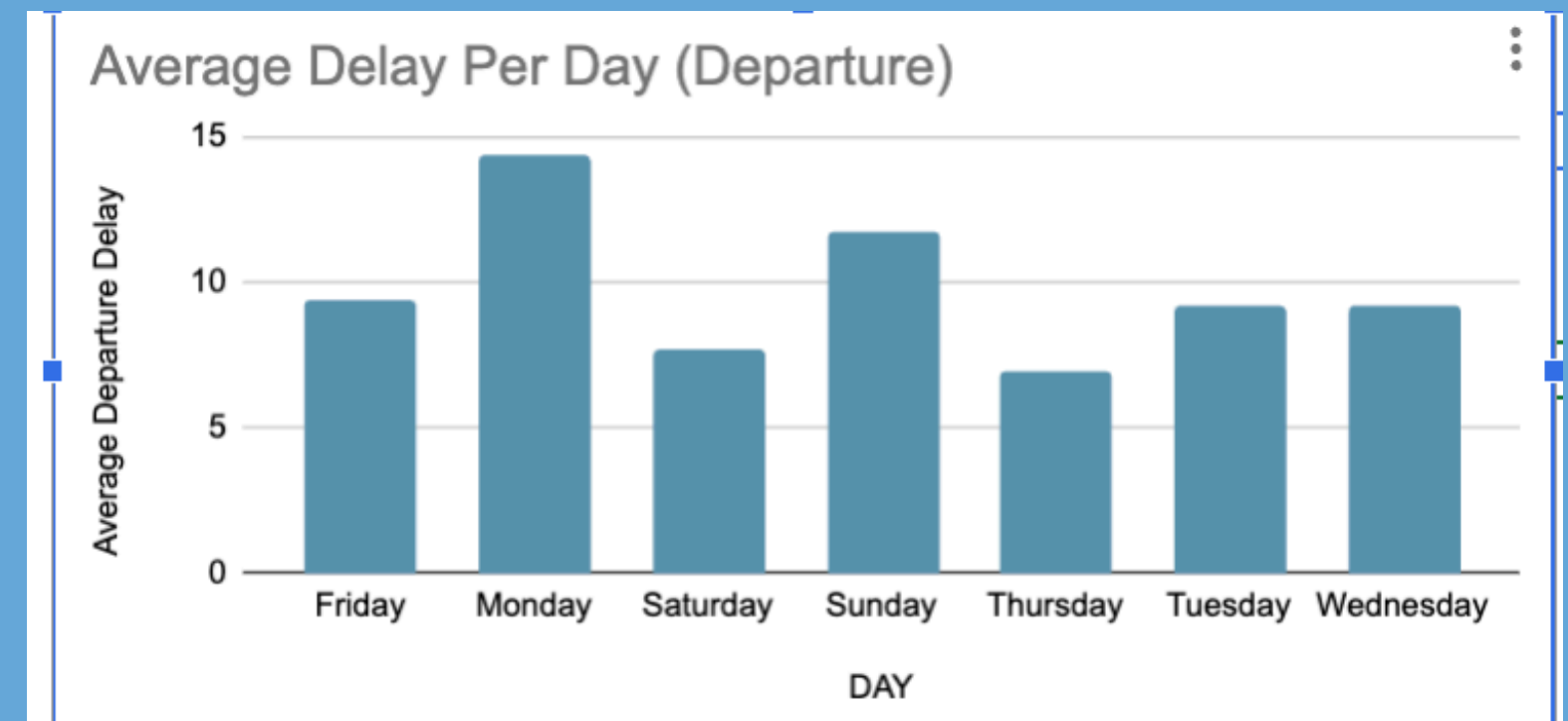
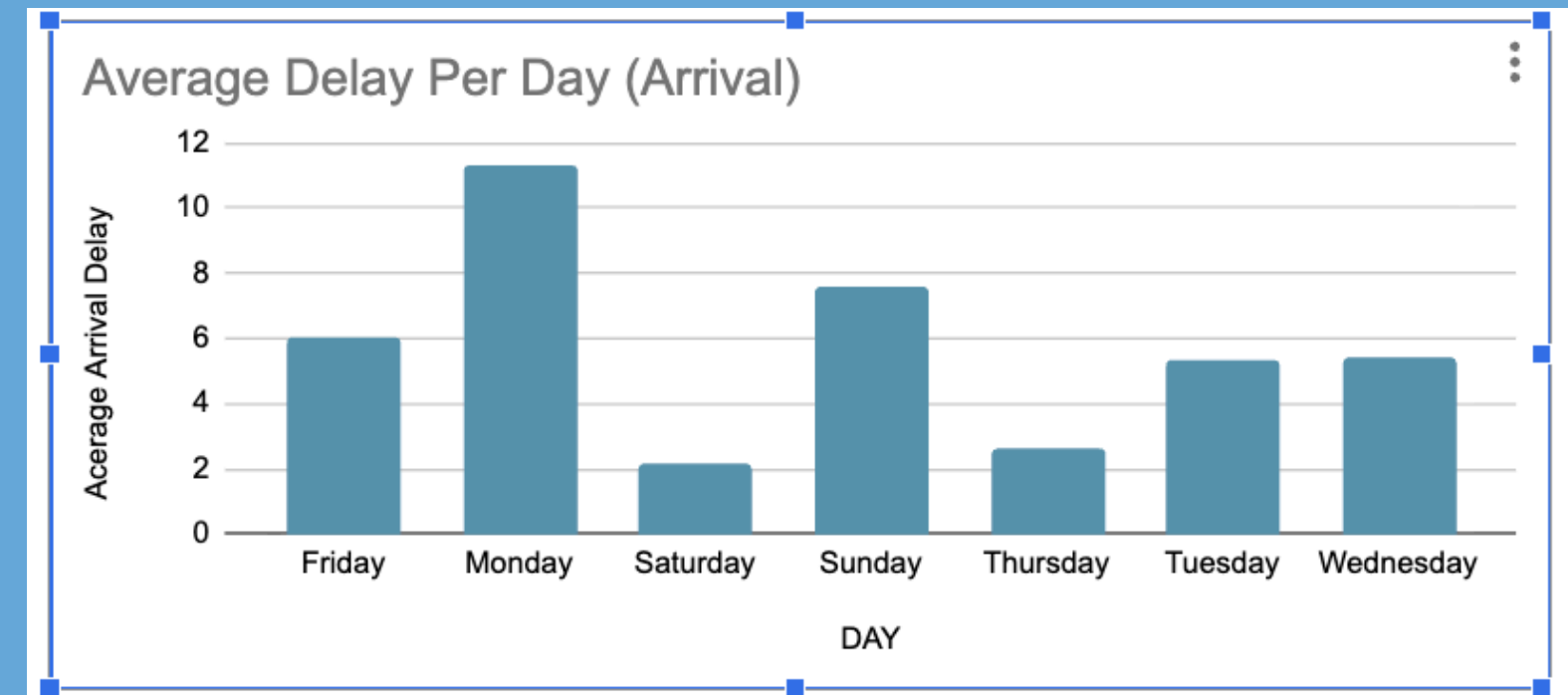
- Delay varies significantly across airlines
- High traffic ≠ high delay
- Small subset drives majority of delay



# Day-of-Week Analysis

- Monday highest delay
- Midweek stable
- Structured demand pattern

**\*Peak demand drives congestion-driven delay escalation.**

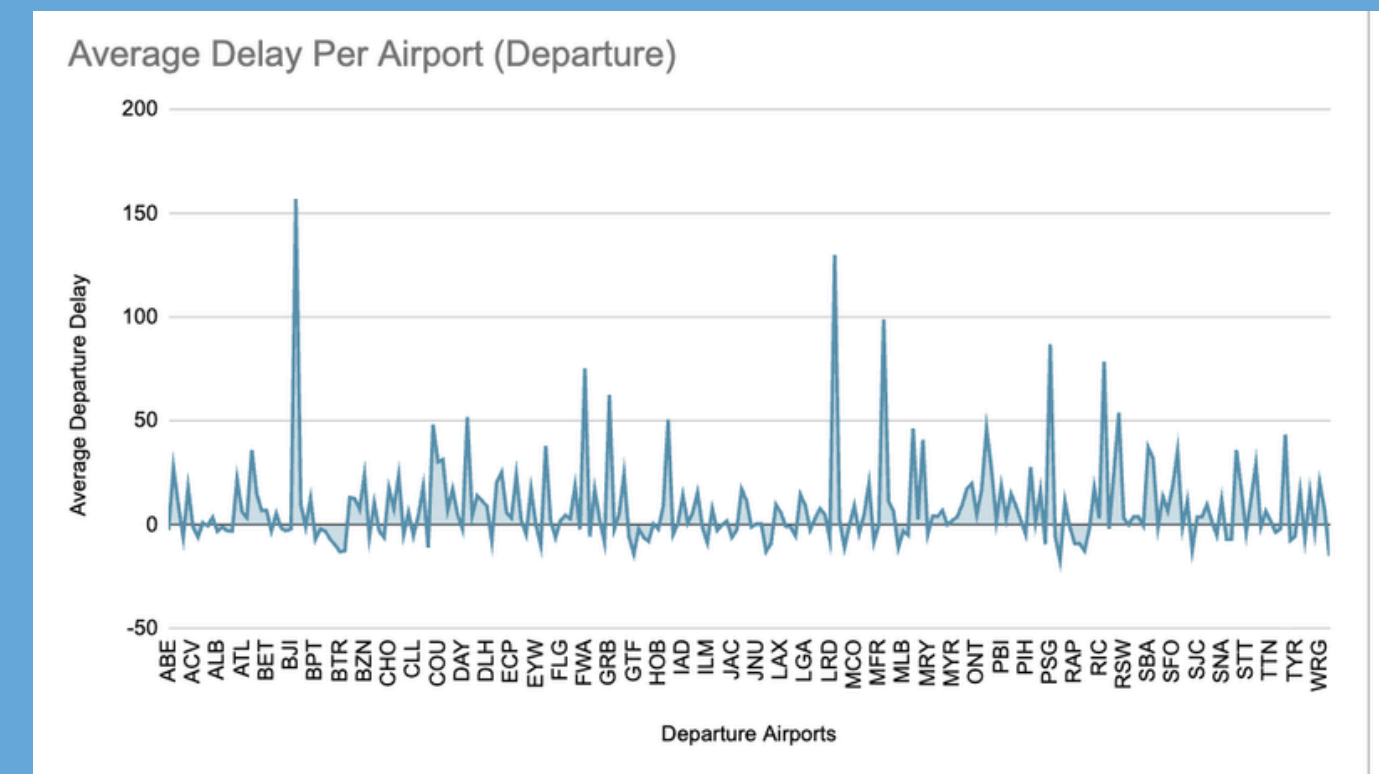
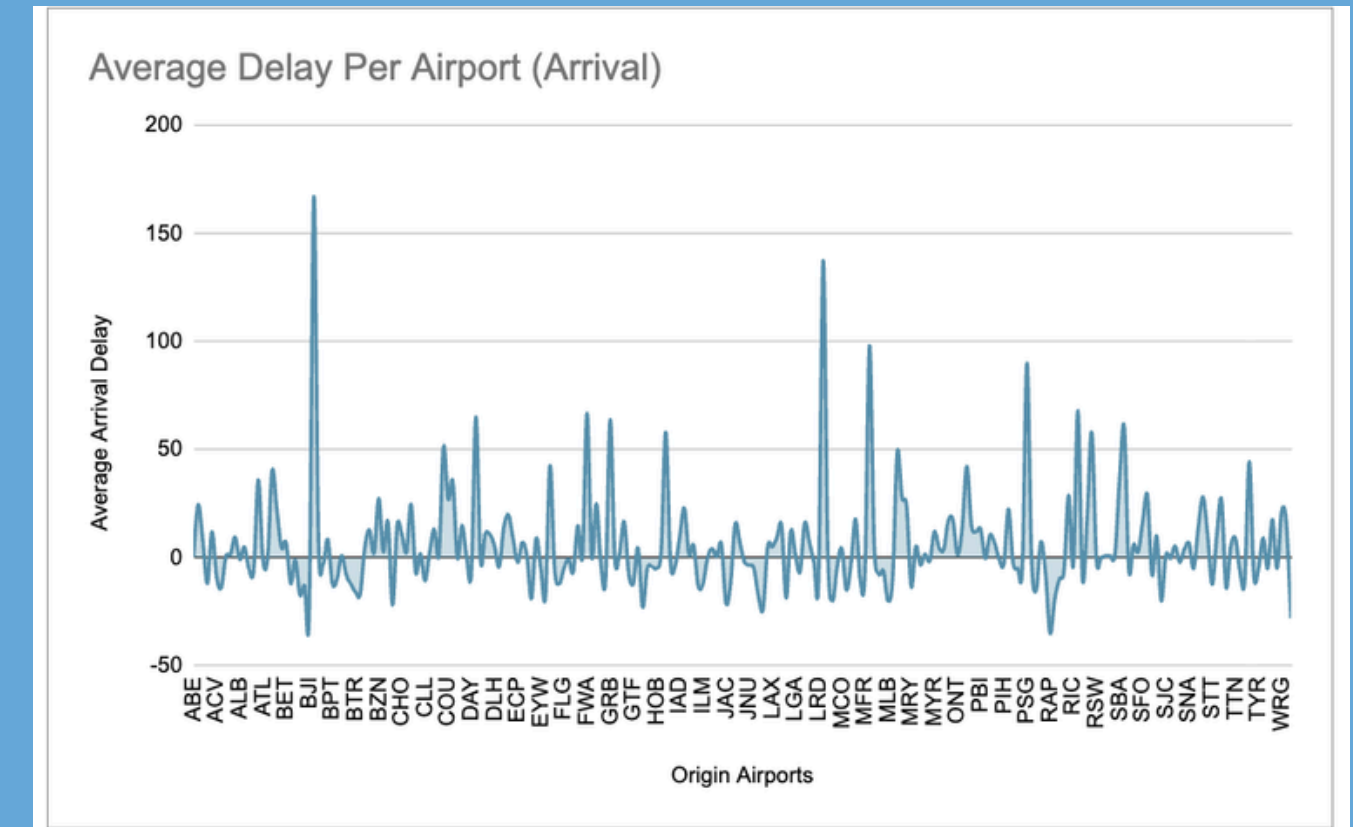




# Airport Performance

- Delay clustering at specific airports
- Hub airports act as bottlenecks
- Arrival & departure delays correlated

**\*Delay concentration nodes create cascading network impact.**





# **Integrated Insight**

- **Airline inefficiency**
- **Weekday congestion**
- **Airport bottlenecks**

**Delays are structural, not random.**

# Strategic Recommendations

Airline operational audit

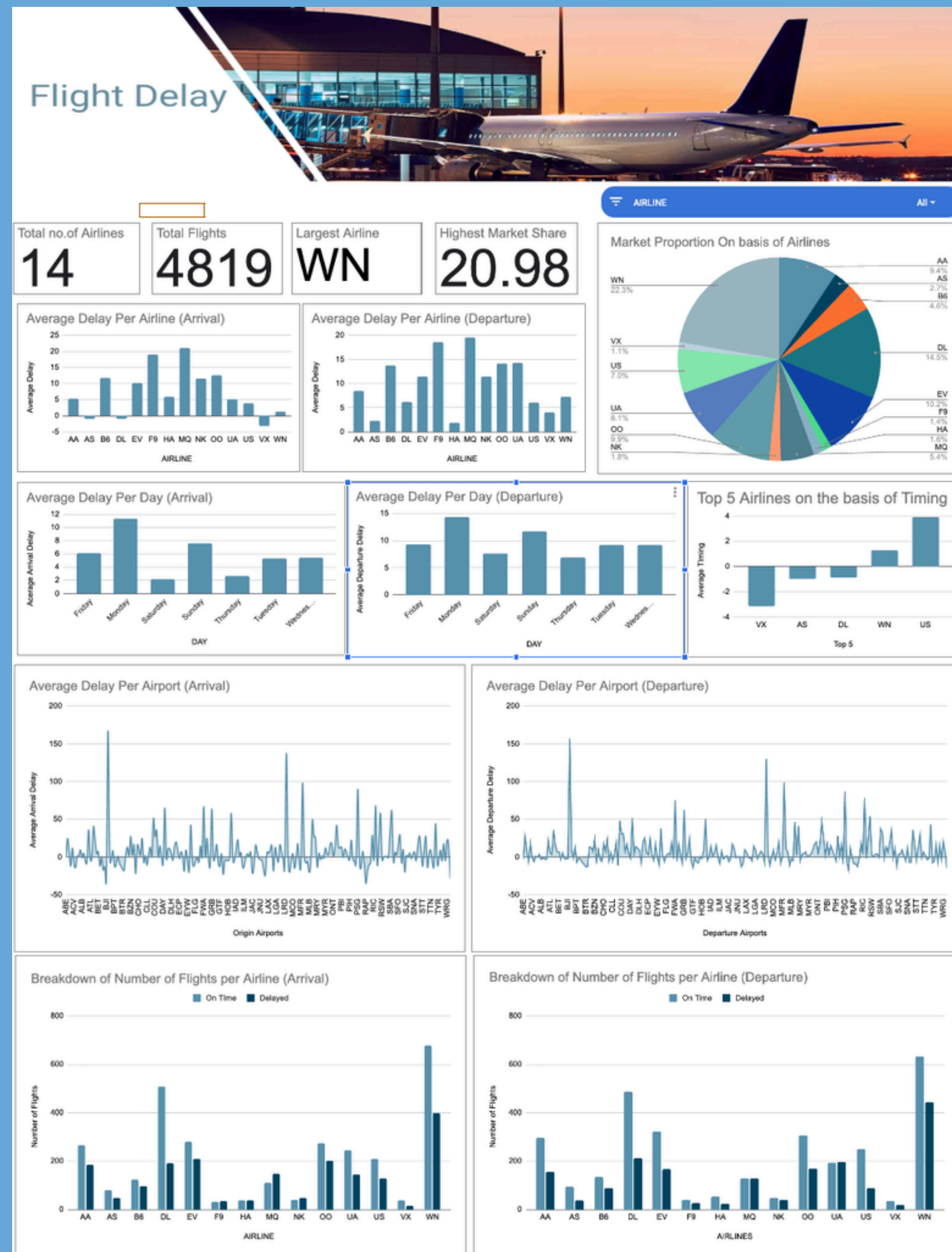
Peak weekday  
buffer scheduling

Hub airport  
resource optimization

Continuous delay  
monitoring system



# Dashboard Walkthrough



The dashboard provides a layered performance view starting with executive KPIs, followed by airline and weekday benchmarking, and finally airport-level delay concentration.

An interactive airline filter enables dynamic segmentation across all visuals.

This structure allows decision-makers to quickly identify underperforming carriers, peak congestion days, and operational bottlenecks within the network.

# Conclusion

Delays are concentrated, measurable, and structured.

Targeted interventions outperform generic policies.

Dashboard enables continuous performance monitoring.